October 28, 2026

Editor
The R Journal

Dear Editor,

Please consider my article titled “Fast and Memory-Light Evaluation of Density-Based Clusterings with the DISCO Index in R” for publication in the R Journal.

DISCO is a recent internal cluster validity index for density-based clustering (Beer et al., ICLR 2026) and the first to evaluate the quality of noise labels. Its definition relies on distances between all pairs of points, so direct implementations, including the first CRAN release of my package discoCVI, need several n x n matrices and are limited to a few thousand points.

The article shows that every quantity DISCO requires can be obtained from the single-linkage merge tree of the mutual-reachability minimum spanning tree in one pass, which gives an exact algorithm with memory linear in the number of points. It describes the design of the package and a typical workflow (also provided as a vignette), the implementation in discoCVI 0.2.0, its test suite, a correction of the score of singleton clusters to match the published definition, a comparison with the original Python implementation (identical scores, up to nine times faster and with a fraction of the memory), and a benchmark of time, memory and energy consumption. I believe the article is of interest to R Journal readers working on clustering and model selection, and to those concerned with the computational and environmental cost of statistical software.

The package is on CRAN (version 0.2.0, with the implementation described in the article). Version 0.2.1, which only adds the workflow vignette described in the article, was submitted to CRAN on 28 October 2026 and is under review; it is already available on GitHub (https://github.com/aminentezari/discoCVI).

The manuscript has not been published or submitted elsewhere.

Regards,

Amin Entezari
Università di Milano-Bicocca, Milan, Italy
amin_entezari@outlook.com